

UNIT V: ELECTROMAGNETIC WAVES

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Session Objectives

- To understand electromagnetic wave propagation and characterize the parameters.

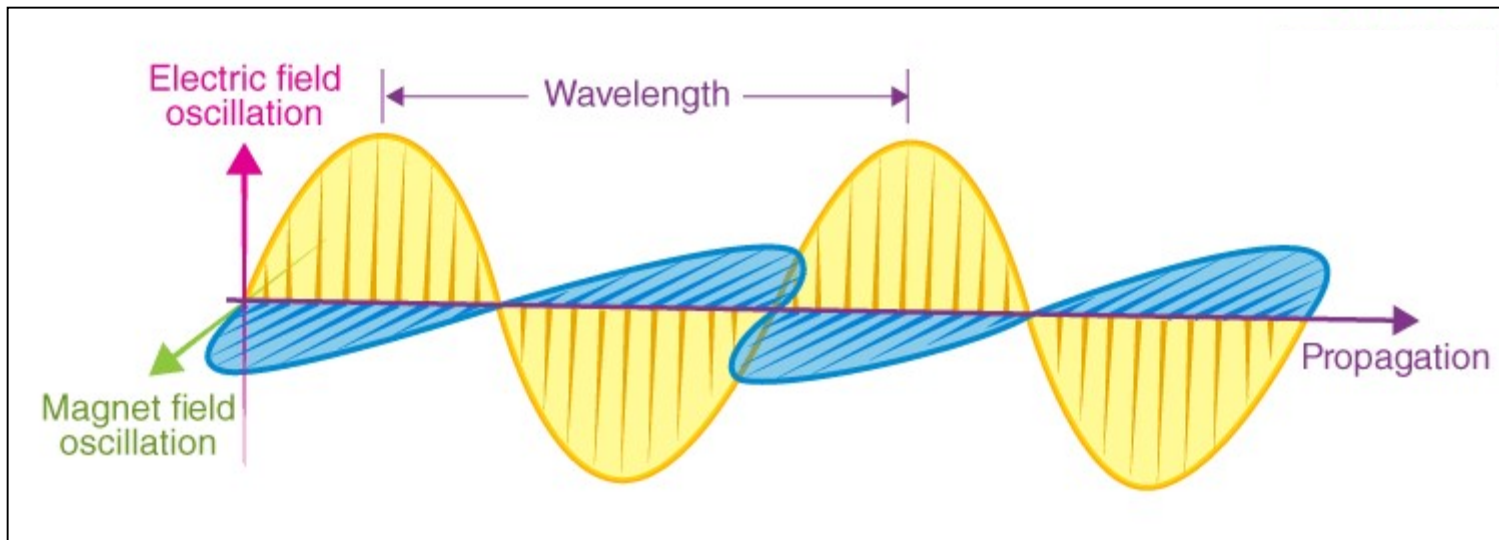
Session Outcomes

- Students will be able to illustrate the electromagnetic wave propagation and corresponding parameters. (CO5)

Session Outline

- Equation of Electromagnetic Wave in Free-Space
- Concept of Lossless Medium and Lossy Medium
- Attenuation constant, Phase constant, Propagation Constant
- General Electromagnetic Wave Equation in Lossless and Lossy Mediums
- Skin Effect and Skin Depth
- Intrinsic Impedance
- Poynting Vector and Poynting's Theorem
- Calculation of Power Density, Average Power Density, and Average Power of Electromagnetic Waves

Electromagnetic Wave



NOTE: The Direction of Propagation of an Electromagnetic Wave is Perpendicular to both Electric Field and Magnetic Field.

Maxwell's equations in free-space

maxwell's equation in free-space takes the following form:

$$\textcircled{1} \vec{\nabla} \cdot \vec{D} = \vec{\nabla} \cdot \vec{E} = 0$$

$$\textcircled{2} \vec{\nabla} \cdot \vec{B} = \vec{\nabla} \cdot \vec{H} = 0$$

$$\textcircled{3} \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$\textcircled{4} \vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Constraints in free-space:

- $\rho = 0$
- $\sigma = 0$, so $\vec{J}_c = 0$
- $\vec{B} = \mu_0 \vec{H}$
- $\vec{D} = \epsilon_0 \vec{E}$

In order to solve maxwell's equation, curl of eq. ~~3~~ $\textcircled{3}$ and eq $\textcircled{4}$ are taken.

$\therefore$ By taking curl of eq $\textcircled{3}$ we can get the solution of maxwell's equation for $\vec{E}$.

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left(-\mu_0 \frac{\partial \vec{H}}{\partial t} \right)$$

$$a, \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H})$$

$$a, \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{\partial \vec{E}}{\partial t} \right)$$

[using vector triple product rule in the l.h.s. and using eq $\textcircled{4}$ in r.h.s.]

$$a, -\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad [\text{using } \textcircled{2} \text{ and } \textcircled{1}]$$

$$a, \boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}} \quad \text{--- } \textcircled{5}$$

This is the ~~solution~~ of wave equation for $\vec{E}$ in free-space.

Similarly, by taking curl of eq (4) we can get the solution of Maxwell's equations for $\vec{H}$.

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = \epsilon_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{E})$$

$$\therefore \vec{\nabla} (\vec{\nabla} \cdot \vec{H}) - \nabla^2 \vec{H} = \epsilon_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{E})$$

[using the Vector triple product rule in the L.H.S.]

$$\therefore -\nabla^2 \vec{H} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}$$

[using eq (2) in the L.H.S. and eq (3) in the R.H.S.]

$$\therefore \boxed{\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}} \quad \text{--- (6)}$$

This is the wave equation for $\vec{H}$ in free-space

Therefore, combining eq (5) and eq (6), a general wave equation for electromagnetic wave in free-space can be written as,

$$\boxed{\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}} \quad \text{--- (7)}$$

where $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$= \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}}}$$

$$= 2.998 \times 10^8$$

$$\approx 3 \times 10^8 \text{ m/s}$$

$$= c \text{ (speed of light)}$$

- Therefore, electric ~~wave~~ ^{field} $\vec{E}$ and magnetic field $\vec{H}$ propagate ~~is~~ as wave in free-space with the speed of light ($c = 3 \times 10^8$ m/s).
- Also, this suggests that the light is an ~~elect~~ electro-magnetic wave.

~~XXXXXXXXXXXXXXXXXXXX~~

Propagation constant, Attenuation constant, Phase constant

$$\nabla^2 \vec{E} = \frac{1}{\mu^2} \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (5)} \quad \left| \quad \nabla^2 \vec{H} = \frac{1}{\mu^2} \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{--- (6)}\right.$$

(This can be written as, This can be written as,

$$\nabla^2 \vec{E} = \gamma^2 \vec{E} \quad \text{--- (8)} \quad \left| \quad \nabla^2 \vec{H} = \gamma^2 \vec{H} \quad \text{--- (9)}\right.$$

Where, γ is the propagation constant.

Propagation constant γ is given by,

$$\gamma = \alpha + j\beta \quad \text{--- (10)}$$

$\swarrow$ $\downarrow$ $\searrow$
 Propagation Attenuation Phase
 constant constant constant

Propagation constant is given as,

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \quad \text{--- (11)}$$

where ω is the angular velocity
 μ is the permeability
 σ is the conductivity
 ϵ is the permittivity

~~Similarly~~, Solving this equation we get,

$$\text{Propagation constant } \alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\left[\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} - 1\right]} \quad \text{--- (12)}$$

$$\text{Phase Constant } \beta = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\left[\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} + 1\right]} \quad \text{--- (13)}$$

Values of α and β in loss-less & lossy mediums

Loss-less medium : A medium in which there is no loss of electromagnetic wave signal during its propagation is called a loss-less medium.

There is no free charge carriers in a lossless medium, therefore conductivity, $\sigma = 0$.

$\therefore$ Putting $\sigma = 0$ in eq. (12) and (13),

$$\begin{aligned}\alpha &= \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\left[\sqrt{1+0} - 1\right]} \\ &= \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{1-1} \\ &= 0\end{aligned}$$

$$\begin{aligned}\beta &= \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\sqrt{1+0} + 1} \\ &= \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{1+1} \\ &= \omega \sqrt{\mu\epsilon}\end{aligned}$$

Lossy medium? A medium in which there is loss of electromagnetic wave signal during its propagation is called a lossy medium.

The losses in the lossy medium ~~is due~~ occurs due to the absorption of energy of the electromagnetic wave by the charge-carriers present in the lossy medium. For a lossy medium, ~~conductivity~~ conductivity, σ is very high.

Therefore, putting $\sigma \gg \omega \epsilon$ in (12) & (13),

$$\alpha = \omega \sqrt{\frac{\mu \epsilon}{2}} \sqrt{\sqrt{\frac{\sigma^2}{\omega^2 \epsilon^2}} - 1}$$

$$= \omega \sqrt{\frac{\mu \epsilon}{2}} \sqrt{\frac{\sigma}{\omega \epsilon}}$$

$$= \sqrt{\frac{\omega \mu \sigma}{2}}$$

$$\beta = \omega \sqrt{\frac{\mu \epsilon}{2}} \sqrt{\sqrt{\frac{\sigma^2}{\omega^2 \epsilon^2}} + 1}$$

$$= \omega \sqrt{\frac{\mu \epsilon}{2}} \sqrt{\frac{\sigma}{\omega \epsilon}}$$

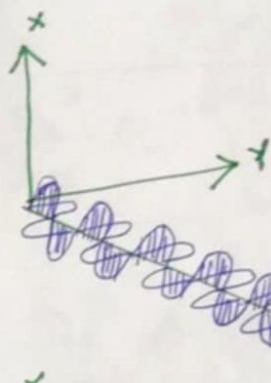
$$= \sqrt{\frac{\omega \mu \sigma}{2}}$$

	α	β
Loss-less medium	0	$\omega \sqrt{\mu \epsilon}$
Lossy medium	$\sqrt{\frac{\omega \mu \sigma}{2}}$	$\sqrt{\frac{\omega \mu \sigma}{2}}$

EM waves in loss-less and lossy mediums

Loss-less medium

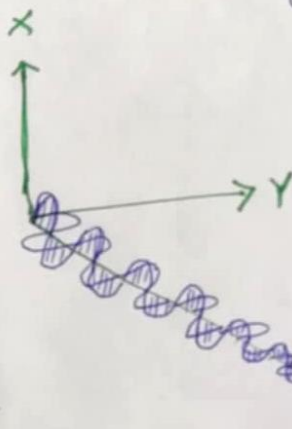
There is no decay in the signal.



$\vec{E}$ in x-plane
 $\vec{H}$ in y-plane
 z is the direction of propagation of the electromagnetic wave.

Lossy medium

There is an exponential decay in the energy or strength of the signal.



$\vec{E}$ in x-plane
 $\vec{H}$ in y-plane
 z is the direction of propagation of the electromagnetic wave.

EM wave equations in free-space/loss-less medium

$$\vec{E} = E_0 \cos(\omega t - \beta z) \hat{i} \quad \vec{H} = H_0 \cos(\omega t - \beta z) \hat{j}$$

EM wave equations in other medium

$$\vec{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{i} \quad \vec{H} = H_0 e^{-\alpha z} \cos(\omega t - \beta z) \hat{j}$$

[The negative sign before βz indicates the propagation is along positive z-axis]

where, ω = angular velocity, t = time,

α = attenuation constant, β = phase constant,

z = direction of propagation of e.m. wave,

$\hat{i}$ = unit vector along x-axis (i.e. along $\vec{E}$).

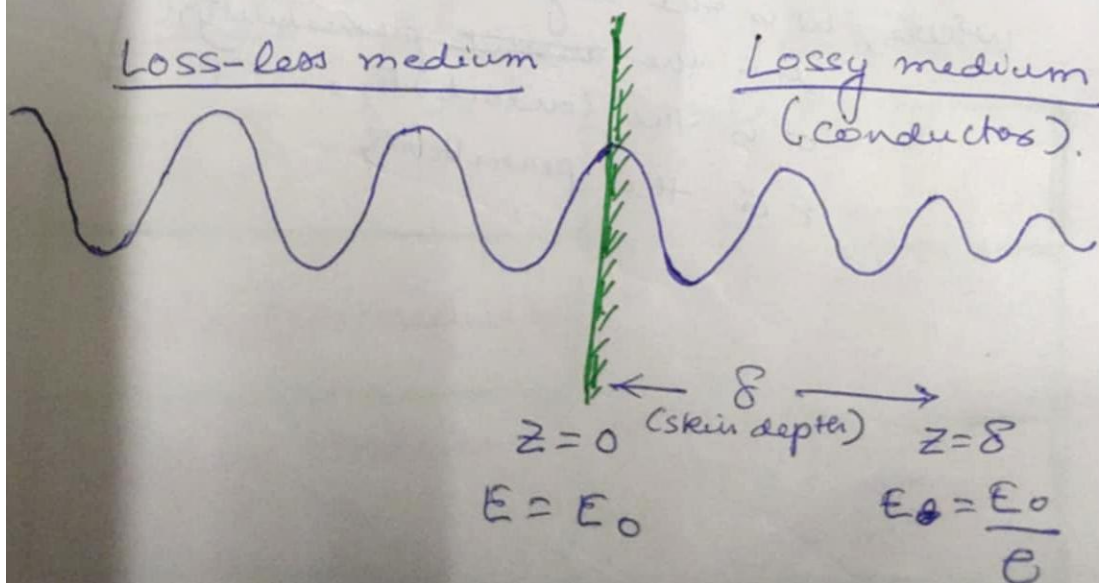
$\hat{j}$ = unit vector along y-axis (i.e. along $\vec{H}$).

Skin effect and skin depth

Skin effect: ~~not~~ As the electromagnetic wave enters ~~inside~~ in a lossy medium (for e.g. - conductor), ~~there~~ there is an exponential decay or degradation or loss in the amplitude or strength of the wave or signal.

The signal strength is highest at the skin or surface and as it propagates inside, there will be loss due to the presence of free charge carriers.

This phenomenon is known as skin-effect.



Skin depth : Skin depth is defined as the depth or distance from the skin or surface at which the electromagnetic wave or signal attenuates to $\frac{1}{e}$ times of the signal amplitude at the surface.

Expression for skin-depth, δ

At the surface of a conductor $z=0$.

$\therefore$ Putting $z=0$,

the signal strength or amplitude at surface is

$$E = E_0 e^{-\alpha \cdot 0} = E_0 \frac{1}{e^0} = E_0$$

Now, at skin-depth, $z=\delta$,

$\therefore$ ~~The signal~~ we know, the signal strength at the skin-depth is $\frac{E_0}{e}$.

Therefore,

$$\frac{E_0}{e} = E_0 e^{-\alpha \delta}$$

$$\text{or, } e^{-1} = e^{-\alpha \delta}$$

$$\text{or, } \alpha \delta = 1 \quad [\text{taking logarithm on both sides}]$$

$$\text{or, } \boxed{\delta = \frac{1}{\alpha}}$$

Therefore,

$$\text{Skin-depth} = \frac{1}{\text{Attenuation constant}}$$

Now, for loss-less medium there will be no losses ($\alpha=0$).

For lossy medium, $\alpha = \sqrt{\frac{\omega \mu \sigma}{2}}$

Therefore, skin depth for lossy medium

$$\delta_{\text{lossy}} = \frac{1}{\alpha} = \frac{1}{\sqrt{\frac{\omega \mu \sigma}{2}}} = \sqrt{\frac{2}{\omega \mu \sigma}}$$

where, ω is the angular velocity,
 μ is the permeability,
 σ is the conductivity.

Intrinsic Impedance

The intrinsic impedance refers to the inherent resistance ^{offered by the medium} to the flow of the electromagnetic wave within a material or medium.

The intrinsic impedance η is given by the ratio of ^{the} electric field intensity to the magnetic field intensity.

It has a unit of Ohm (Ω).

$$\eta = \frac{E}{H}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

where, ω is the angular velocity,
 μ is the ~~absolute~~ permeability,
 σ is the conductivity,
 ϵ is the permittivity.

Values of η in loss-less & lossy mediums

Loss-less medium: $\sigma = 0$, and there is no loss of a signal.

$$\therefore \eta = \sqrt{\frac{j\omega\mu}{j\omega\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} \quad \Omega$$

for free-space,

$$\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}} = 377 \text{ or } 120\pi \quad \Omega$$

Lossy medium: $\sigma \gg 1$, and there will be loss of signal in the medium due to the presence of charge carriers.

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} \quad \Omega$$

	$\eta (\Omega)$
Loss-less medium	$\eta = \sqrt{\frac{\mu}{\epsilon}}$
Lossy medium	$\eta = \sqrt{\frac{j\omega\mu}{\sigma}}$

Poynting's Theorem and Poynting Vector

$$\boxed{\vec{P} = \vec{E} \times \vec{H}} \quad \text{W/m}^2$$

$\vec{P}$ is the Poynting vector which denotes the power density. [In some books, you will find $\vec{P}$ as $\vec{S}$, both are same.]

Unit of $\vec{P}$: unit of $\vec{E}$ $\times$ unit of $\vec{H}$

$$= \frac{\text{V}}{\text{m}} \times \frac{\text{A}}{\text{m}}$$
$$= \text{W/m}^2$$

• Calculation of Power density (W/m^2) :

$$\vec{P} = \vec{E} \times \vec{H} \quad \text{--- (i)}$$

where, electric field intensity

$$\vec{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z + \theta_1) \hat{i} \quad \text{--- (ii)}$$

$$\vec{H} = H_0 e^{-\alpha z} \cos(\omega t - \beta z + \theta_2) \hat{j} \quad \text{--- (iii)}$$

Combining (i), (ii), and (iii),

$$\vec{P} = E_0 e^{-\alpha z} \cos(\omega t - \beta z + \theta_1) \hat{i} \\ \times H_0 e^{-\alpha z} \cos(\omega t - \beta z + \theta_2) \hat{j}$$

$$\text{or, } \vec{P} = E_0 H_0 e^{-2\alpha z} \cos(\omega t - \beta z + \theta_1) \cos(\omega t - \beta z + \theta_2) \hat{k}$$

$$[\because \hat{i} \times \hat{j} = \hat{k}]$$

$$\text{or, } \vec{p} = \frac{E_0 H_0 e^{-2\alpha z}}{2} \left[2 \cos(\omega t - \beta z + \theta_1) \cos(\omega t - \beta z + \theta_2) \right] \hat{k}$$

$$\text{a, } p = \frac{E_0 H_0 e^{-2\alpha z}}{2} \left[\cos(2\omega t - 2\beta z + \theta_1 + \theta_2) + \cos(\theta_1 - \theta_2) \right] \hat{k} \quad \text{--- (iv)}$$

$$\left[\text{using } 2 \cos A \cos B = \cos(A+B) + \cos(A-B) \right]$$

• Calculation of Average Power Density (W/m²):

$$\text{① } P_{\text{avg}} = \frac{1}{T} \int_0^T p \, dt$$

$$\text{a, } P_{\text{avg}} = \frac{1}{T} \int_0^T \frac{E_0 H_0 e^{-2\alpha z}}{2} \left[\cos(2\omega t - 2\beta z + \theta_1 + \theta_2) + \cos(\theta_1 - \theta_2) \right] dt$$

$$\text{e, } P_{\text{avg}} = \frac{1}{T} \cdot \frac{E_0 H_0 e^{-2\alpha z}}{2} \left[\int_0^T \cos(2\omega t - 2\beta z + \theta_1 + \theta_2) dt + \int_0^T \cos(\theta_1 - \theta_2) dt \right]$$

$$\text{a, } P_{\text{avg}} = \frac{1}{T} \cdot \frac{E_0 H_0 e^{-2\alpha z}}{2} \left[0 + \cos(\theta_1 - \theta_2) T \right]$$

$$\text{e, } P_{\text{avg}} = \frac{E_0 H_0 e^{-2\alpha z}}{2} \cos(\theta_1 - \theta_2)$$

$$\text{e, } \boxed{P_{\text{avg}} = \frac{E_0 H_0}{2} e^{-2\alpha z} \cos \theta_\eta} \quad \left[\text{where, } \theta_\eta \text{ is the intrinsic impedance angle.} \right]$$

$$\left[\because \eta = \frac{E}{H} = \frac{E_0 \angle \theta_1}{H_0 \angle \theta_2} = \frac{E_0}{H_0} \angle (\theta_1 - \theta_2) = \frac{E_0}{H_0} \angle \theta_\eta \right]$$

Also, $\eta = \frac{E_0}{H_0}$ [where η is the intrinsic impedance in Ω .]

$$\therefore \boxed{P_{avg} = \frac{E_0^2}{2\eta} e^{-2\alpha z} \cos \theta \eta} \quad \left[\because H_0 = \frac{E_0}{\eta} \right]$$

also,

$$\boxed{P_{avg} = \frac{\eta H_0^2}{2} e^{-2\alpha z} \cos \theta \eta}$$

Calculation of Average Power (W):

Average Power = Average Power Density $\times$ Area

Problems

1. Calculate the velocity of an electromagnetic wave propagating through a medium with relative permittivity 3 and relative permeability 1.

Given data, $\epsilon_r = 3$ and $\mu_r = 1$.

The velocity of the electromagnetic wave,

$$v = \frac{1}{\sqrt{\mu \epsilon}}$$

$$= \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$

$$= \frac{1}{\sqrt{\mu_0 \epsilon_0} \sqrt{\mu_r \epsilon_r}}$$

$$= \frac{1}{\sqrt{\mu_0 \epsilon_0}} \times \frac{1}{\sqrt{\mu_r \epsilon_r}}$$

$$= 3 \times 10^8 \times \frac{1}{\sqrt{1 \times 3}} \quad \left[\because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s} \right]$$

$$= \sqrt{3} \times 10^8$$

$$= 1.732 \times 10^8 \text{ m/s}$$

2. The time-domain expression of a magnetic field wave travelling in a non-magnetic medium is given by

$$H(x,t) = \frac{1}{2} \cos(6\pi \times 10^8 t - 10x) \hat{k} \text{ A/m.}$$

Find the following [for a lossless medium]:

- the direction of wave propagation,
- the dielectric constant,
- the intrinsic impedance,
- the time-domain expression of electric field,
- the average power density.

Given data, for non-magnetic medium $\mu_r = 1$

Comparing the given equation for general magnetic wave equation for lossless medium

we get, i.e. $H = H_0 \cos(\omega t - \beta x) \hat{k}$

we get, $H_0 = \frac{1}{2}$, $\omega = 6\pi \times 10^8$, $\beta = 10$, and

- the direction of wave propagation is along the positive x -axis,

(b) now, for lossless medium,

$$\beta = \omega \sqrt{\mu \epsilon} = 10 \quad (\text{given}).$$

$$\therefore \omega \sqrt{\mu \epsilon} = 10 \quad \omega \sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r} = 10$$

$$\text{or, } 6\pi \times 10^8 \sqrt{1 \cdot \epsilon_r} \cdot \sqrt{\mu_0 \epsilon_0} = 10$$

$$\text{or, } \frac{6\pi \times 10^8}{3 \times 10^8} \sqrt{1 \cdot \epsilon_r} = 10 \quad \left[\because \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{3 \times 10^8}{\text{m/s}} = c \right]$$

$$\therefore \epsilon_r = \left(\frac{10}{2\pi} \right)^2 = 2.533$$

(c) For loss-less medium, intrinsic impedance,

$$\begin{aligned}\eta &= \sqrt{\frac{\mu}{\epsilon}} \\ &= \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} \\ &= \sqrt{\frac{4\pi \times 10^{-7} \times 1}{8.854 \times 10^{-12} \times 2.533}} \\ &= 236.71 \Omega\end{aligned}$$

(d) The ^{general} ~~time~~ time domain expression for electric field will be,

$$E(x, t) = \frac{E_0}{2} \cos(\omega t - \beta x) \hat{j} \text{ V/m}$$

Here, $\omega = 6\pi \times 10^8$ and $\beta = 10$.

$$\text{Now, } \eta = \frac{E_0}{H_0}$$

$$\therefore 236.71 = \frac{E_0}{\frac{1}{2}}$$

[as the direction of propagation is along x-axis and $\vec{H}$ is along z-axis ($\hat{k}$), so $\vec{E}$ will be along y-axis ($\hat{j}$).]

$$\therefore E_0 = 236.71 \times \frac{1}{2} = 118.355$$

Therefore, the time domain expression for electric field will be,

$$E(x, t) = 118.355 \cos(6\pi \times 10^8 t - 10x) \hat{j} \text{ V/m.}$$

(e) Average power density, $P_{avg} = \frac{E_0 H_0}{2} e^{-2\alpha z} \cos \theta_n$

$$= \frac{E_0^2}{2\eta} e^{-2\alpha z} \cos \theta_n$$

$$= \frac{\eta H_0^2}{2} e^{-2\alpha z} \cos \theta_n$$

for loss-less medium, $\alpha=0$,

also, $\theta_n = 0$

~~$$P_{avg} = \frac{\eta H_0^2}{2} \cos \theta_n$$~~

$$\therefore P_{avg} = \frac{E_0 H_0}{2} = \frac{E_0^2}{2\eta} = \eta \frac{H_0^2}{2}$$

$$= \frac{236.71 \times \left(\frac{1}{2}\right)^2}{2} = 29.589 \text{ W/m}^2$$

Test Your Understanding

1. What is an electromagnetic wave? Calculate the velocity of an electromagnetic wave in free space.
2. Derive a mathematical expression for electromagnetic wave in free-space.
3. State 2 differences between lossless medium and lossy medium. What causes the losses of an electromagnetic wave in a lossy medium.
4. Write Poynting's theorem. What is Poynting vector? Mention it's unit.
5. Explain skin-effect. Derive a mathematical relation between the skin depth and attenuation constant. Then, find the expression for skin-depth in lossy medium.
6. Derive expressions for attenuation constant, phase constant, and propagation for lossless and lossy mediums.
7. What is the intrinsic impedance? Mention it's unit. Find expressions for intrinsic impedance for lossless and lossy mediums. Then, calculate the value of intrinsic impedance of free-space.
8. Determine the wavelength, attenuation constant, and phase constant of an electromagnetic wave traveling in a free space with a frequency of 500 MHz.
9. Calculate the skin depth of a copper conductor for an electromagnetic wave propagating with a frequency of 60 Hz. The conductivity of copper is 5.8×10^7 S/m, and its relative permeability is approximately 1.

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Thank you !

